(ii) P (less than 56 marks) $= P(X < 56)$

$$= P\left(Z < \frac{56-60}{5}\right)$$

$$= P(Z < -0.8)$$

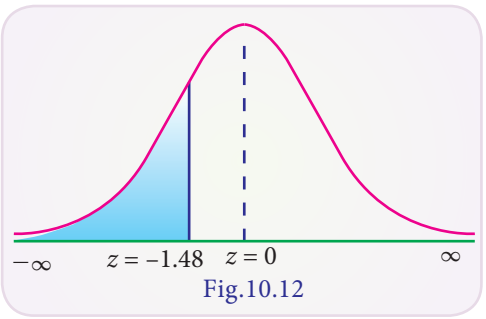


Fig.10.12

$= P(-\infty < Z < 0) - P(-0.8 < Z < 0)$

$$= 0.5000 - P(0 < Z < 0.8)$$

$$= 0.5000 - 0.2881$$

$$= 0.2119$$

$\therefore$ Number of students scored less than 50 marks $= 0.2119 \times 100$

$= 21.19\%$

(iii) P (between 45 and 65 marks) $= P(45 < x < 65)$

$$= P\left(\frac{45-60}{5} < z < \frac{65-60}{5}\right)$$

$$= P(-3 < z < 1)$$

$$= P(0 < Z < 3) + P(0 < Z < 1)$$

$$= 0.4986 + 0.3413$$

$$= 0.8399$$

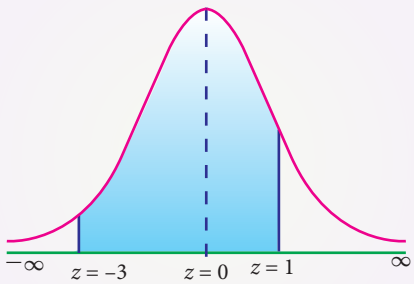


Fig.10.13

Number of students scored between 45 and 60 marks = 0.8399 x 100= 83.99%

Example 10.32

X has Normal distribution with mean 2 and standard deviation 3. Find the value of the variable x such that the probability of the interval from the mean to that value is 0.4115.

Solution:

Given $\mu = 2,\ \sigma = 3, z = \frac{x-\mu}{\sigma} = \frac{x-2}{3}$

Let $Z_1 = \frac{x_1 - 2}{3}$

We have $P(\mu < x < x_1) = 0.4115$

$P(0 < Z < z_1) = 0.4115$

But $P(0 < Z < 1.35) = 0.4115$ (from the Normal Probability table)

$\therefore \quad Z_1 = 1.35 \therefore \frac{x_1 - 2}{3} = 1.35$ (or) $x_1 = (1.35) \times 3 + 2 = 6.05$